

From Proposal to Thesis: Updated Econometric Design and Role of K/L

1. Purpose of this Memo

This memo explains how the **actual thesis** will differ from the **original proposal**, and sets out a clear, internally consistent econometric design built around:

1. Two parallel **not-yet-treated (NYT) event-study models**:
2. **Model 1A**: Reform $\rightarrow$ log labor productivity, $\ln(LP_{i,t})$.
3. **Model 1B**: Reform $\rightarrow$ log capital deepening, $\ln(K_{i,t}/L_{i,t})$.
4. A separate **elasticity model** for LP with respect to K/L (**Model 2**), and a **mediation-style decomposition** that uses all three models.

The goal is to carry forward the core ideas of the proposal—reforms, productivity, and capital deepening—while dropping an infeasible IV-mediation plan and replacing it with a design that better matches the data and identification environment.

This memo is meant to guide rewriting the proposal text into an actual thesis draft: revising the conceptual narrative, the econometric section, and the planned tables/figures.

2. Original Proposal: What It Said and Why It Needs Revising

2.1 Original goals and questions

The proposal set out three main empirical goals:

1. **Measure the causal impact of Israel's port reforms** (competition from new private terminals and privatization of Haifa legacy) on terminal-level labor productivity (LP).
2. **Understand mechanisms**, particularly the role of capital deepening (K/L) and service quality improvements (wait times, berth congestion, etc.).
3. **Implement an econometric design** that used modern staggered-adoption DiD methods (NYT/event studies à la Callaway–Sant'Anna and Sun–Abraham) and, in a second step, an IV-based mediation strategy following Dippel & Ferrara.

The proposal already formalized a **NYT event-study regression** for $\ln(LP_{i,t})$ with terminal fixed effects and calendar-month fixed effects, for each reform clock. That will remain the basis of Model 1A.

2.2 Original econometric design

The original design can be summarized as:

1. **Primary event-study (NYT) for LP**:

2. Reforms (competition entry, privatization) treated as staggered shocks.
3. Event-time dummies relative to each reform, with $m = -1$ (last pre-reform month) omitted.
4. Not-yet-treated strategy to avoid contamination from already-treated controls.

5. Instrumental-variable mediation using K/L :

6. Construct a measure of capital deepening K/L at the port/terminal level.
7. Use **engineering milestones** (crane arrivals, berth extensions, major infrastructure completions) as instruments for K/L .
8. Run a 2SLS system where the reforms affect LP partly directly and partly via the instrumented mediator K/L .

The ambition was to quantify the indirect effect of reforms on LP operating through capital deepening, using formal IV mediation techniques.

2.3 Problems that emerged

As we began implementing the design and collecting data, several conceptual and practical issues became clear:

1. **Engineering milestones are poor instruments.**
2. They clearly move capital and K/L , but they also **directly change productivity** by reducing waiting times and increasing berth/crane capacity.
3. Exclusion ("only affect LP through K/L ") is not credible.
4. **K/L and LP share labor in the denominator.**
5. Both $LP = Y/L$ and K/L move mechanically when L changes.
6. This makes mediation via IV even harder: even a valid instrument for K could shift LP simply by shifting L .
7. **Data work naturally pushed toward real K measurement, not IVs.**
8. You invested heavily in building a **Path B capital series** from Note-8 PPE data, big-project tables, and a national GFCF deflator.
9. This makes K/L a rich, directly measured object, not a latent variable in need of indirect instrumentation.
10. **Advisor feedback emphasized mechanisms and robustness, not an IV hunt.**
11. You were encouraged to:
 - Obtain real labor data.
 - Check robustness with alternative L proxies.
 - Use stacked event-studies when multiple shocks are close in time.

- Treat capital deepening as a mechanism while being honest about identification.

These considerations motivate a redesign: **drop the IV-mediation idea** and instead use **measured K/L**, with clean event-studies and a transparent mechanism decomposition.

3. New Conceptual Framework

The new framework keeps the same broad story but reorders the roles of the objects:

1. **Reforms are exogenous policy shocks.**
 2. Entry of Bayport (Haifa competition clock).
 3. Entry of HCT (Ashdod competition clock).
 4. Privatization of Haifa legacy (ownership clock).
 5. **Two primary outcomes of interest:**
 6. **Labor productivity** at the terminal-month level, $LP_{i,t}$ (e.g. mix-adjusted TEU per labor hour, in logs $Y_{i,t} = \ln LP_{i,t}$).
 7. **Capital deepening** at the terminal-month level, $K_{i,t}/L_{i,t}$, with $C_{i,t} = \ln(K_{i,t}/L_{i,t})$.
 8. **Capital is carefully measured, not instrumented.**
 9. Path B uses Note-8 PPE flows, a GFCF deflator, and a big-project table to construct an annual real K path for Haifa legacy.
 10. This is then interpolated to monthly K and mapped to terminals (legacy vs new terminals, as feasible).
 11. **K/L is treated as a measured mechanism, not as an instrumented regressor.**
 12. We estimate **how reforms change K/L** over time.
 13. Separately, we estimate **how LP co-moves with K/L** in the panel.
 14. A mediation-style decomposition then quantifies how much of the reform-induced LP change is consistent with capital deepening.
 15. **No single “magic” instrument is required.**
 16. This avoids over-claiming exclusion restrictions that do not fit the port context.
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4. Updated Econometric Design

4.1 Model 1A – NYT Event-Study: Reforms → ln(LP)

For each reform r (Haifa competition, Ashdod competition, Haifa privatization) and its impacted terminals, we continue to estimate the **NYT event-study** used in the proposal, with $\ln(LP_{i,t})$ as the outcome:

$$\ln(LP_{i,t}) = \sum_{m \in \mathcal{M}_r, m/} \beta_m^{Y,r} \mathbf{1}\{M_{i,t}^r = m\} + \gamma_i + \delta_t + \varepsilon_{i,t}^{Y,r}.$$

- i : terminal (or port pre-split),
- t : calendar month,
- $M_{i,t}^r$: event time in months relative to reform r , with $M_{i,t}^r = 0$ in the go-live month, $m = -1$ omitted as the last pre-reform month,
- γ_i : terminal fixed effects,
- δ_t : calendar-month fixed effects,
- Sample restriction: only not-yet-treated or never-treated terminal-months for reform r enter as controls (NYT design).

Window averages such as

$$\beta_{[a,b]}^{Y,r} = \frac{1}{b-a+1} \sum_{m=a}^b \beta_m^{Y,r}$$

summarize the medium-run effect of each reform on LP in months $[a, b]$ (e.g. 1–12, 13–24, full post).

4.2 Model 1B – NYT Event-Study: Reforms → ln(K/L)

We **clone the same specification**, but now take **log capital deepening** as the dependent variable:

$$\ln\left(\frac{K}{L}\right)_{i,t} = \sum_{m \in \mathcal{M}_r, m/} \beta_m^{C,r} \mathbf{1}\{M_{i,t}^r = m\} + \gamma_i + \delta_t + \varepsilon_{i,t}^{C,r}.$$

- The event-time variable, NYT restriction, FE structure, and clustering are identical to Model 1A.
- The coefficients $\beta_m^{C,r}$ measure the dynamic effect of reform r on capital per worker.

This parallel structure yields **two matched event-study paths** for each reform:

- $\{\beta_m^{Y,r}\}$: LP response.
- $\{\beta_m^{C,r}\}$: K/L response.

Thus, the thesis can show, for example, "Two years after Bayport's entry, LP at Haifa legacy is up ~X% and K/L is up ~Z%." Both objects are identified using the same NYT DiD logic.

4.3 Model 2 – Elasticity of LP with Respect to K/L

Next, we estimate a **port-level elasticity** of LP with respect to K/L using the full terminal×month panel (or the NYT subsample for robustness). The baseline specification is:

$$\ln(LP_{i,t}) = \lambda \ln \left(\frac{K}{L} \right)_{i,t} + \gamma_i + \delta_t + u_{i,t}.$$

- λ is the **elasticity of labor productivity with respect to capital per worker**, conditional on terminal and calendar-month fixed effects.
- This is the empirical counterpart of the capital share in a Cobb–Douglas production function, where $\ln(Y/L) = \alpha \ln(K/L) + \ln A$.

We will:

1. Estimate $\hat{\lambda}$ with FE and clustered SEs.
2. Compare $\hat{\lambda}$ to a benchmark **capital-income share** $\bar{\alpha}$ computed from Haifa Port's accounts (value added vs depreciation + imputed return on capital).
3. Optionally estimate a first-difference version $\Delta \ln(LP_{i,t}) = \lambda^{(\Delta)} \Delta \ln(K/L)_{i,t} + \dots$ as a growth-accounting robustness check.

4.4 Mediation-Style Decomposition

Finally, we combine Models 1A, 1B, and 2 in a **mediation-style mechanism analysis**.

For a given reform r and post-reform window $[a, b]$:

1. **Total effect of reform on LP** (from Model 1A):

$$\widehat{\text{TE}}_{[a,b]}^{Y,r} = \beta_{[a,b]}^{Y,r}.$$

1. **Effect of reform on K/L** (from Model 1B):

$$\widehat{\Delta C}_{[a,b]}^r = \beta_{[a,b]}^{C,r}.$$

1. **Elasticity of LP w.r.t. K/L** (from Model 2):

$$\hat{\lambda}.$$

1. **Implied indirect effect via K/L:**

$$\widehat{\text{IE}}_{[a,b]}^r \approx \hat{\lambda} \cdot \widehat{\Delta C}_{[a,b]}^r.$$

1. **Residual (non-capital) component:**

$$\widehat{\text{DE}}_{[a,b]}^r \approx \widehat{\text{TE}}_{[a,b]}^{Y,r} - \widehat{\text{IE}}_{[a,b]}^r.$$

This decomposition is **not IV-based causal mediation**. Instead, it is a **mechanism accounting exercise**:

- We estimate how much K/L actually moves because of the reform.

- We use a production-function-consistent elasticity to translate that K/L change into an LP change.
- We compare this implied LP change to the total effect to see how large a share of the LP gains is quantitatively consistent with capital deepening.

5. Data Construction Under the New Design

5.1 Labor and LP

The LP measure in the proposal is based on mix-adjusted throughput (TEU and tons) and a KEI-style LP proxy. Going forward, the thesis will:

1. **Use actual monthly labor hours** from the Ports Authority (once obtained) to define:

$$LP_{i,t} = \frac{TEU_{i,t}}{Hours_{i,t}} \Rightarrow Y_{i,t} = \ln(LP_{i,t}).$$

2. Maintain a **proxy LP measure** based on available KPIs as a **robustness and backfill option** for periods or terminals where hours are incomplete.
3. Address the advisor's concern that LP and K/L share L by explicitly constructing **alternative L definitions**:
4. Hours vs headcount,
5. Alternative allocation rules across terminals,
6. Possibly wage-bill-based implied hours.

These alternative L series will feed into **alternative definitions of both LP and K/L**, and all main models (1A, 1B, 2) will be re-estimated to check robustness.

5.2 Capital K and K/L

The capital side rests on the **Path B "hybrid" K construction**:

1. **Annual financial backbone:**
2. Use `haifa_financials_raw.tsv` (Note-8 PPE flows, depreciation, disposals) and a national GFCF deflator to build a constant-price PPE series.
3. Apply a **perpetual inventory method (PIM)** with appropriate depreciation (geometric or hyperbolic) to get annual real K for Haifa legacy.
4. **Big-project overlay:**
5. Use `Haifa_big_projects.csv` to identify discrete capital shocks (crane purchases, berth rehabilitations, seafront relocation compensation, Sea Dept sale, etc.).
6. Assign **commissioning months, nominal costs, owners, and asset lives** to each project.

7. Map projects to terminals (legacy berths, Kishon East/West, etc.) and incorporate them as **step changes** in K on top of the smooth PIM path.

8. Monthly interpolation:

9. Interpolate annual K to monthly K (e.g. via Denton or linear interpolation constrained by year-end PIM values).

10. Superimpose project-level steps at the appropriate months.

11. Capital deepening per terminal:

12. Combine monthly K with monthly L to construct $K_{i,t}/L_{i,t}$ and $C_{i,t} = \ln(K_{i,t}/L_{i,t})$ for each terminal.

The result is a **terminal×month panel** with LP , K/L , and reform clocks ready for Models 1A, 1B, and 2.

5.3 Reform Timings and NYT Event-Time Clocks

The reform structure from the proposal is preserved:

- **Haifa competition clock:** Bayport opening date; treated terminals are Haifa legacy terminals; controls are not-yet-treated or never-treated terminals under NYT.
- **Ashdod competition clock:** HCT opening date; parallel structure for Ashdod legacy vs entrant.
- **Haifa privatization clock:** closing date of Haifa Port Company sale; treated unit is Haifa legacy, with suitable controls.

Event-time variables $M_{i,t}^r$ and sets of bins $\mathcal{M}_r$ are already defined in the proposal and will be reused for both $\ln(LP)$ and $\ln(K/L)$. Stacked event-study implementations can be used for robustness where shocks are close in time.

6. Identification, Interpretation, and Robustness

6.1 What remains identified

- **Models 1A and 1B (NYT event-studies)** retain their original identification logic:
- Under parallel trends within the NYT comparison group, the event-time coefficients $\beta_m^{Y,r}$ and $\beta_m^{C,r}$ are consistent estimates of the dynamic treatment effects of reforms on $\ln(LP)$ and $\ln(K/L)$, respectively.
- Terminal FE remove time-invariant heterogeneity; calendar-month FE remove common macro shocks.
- **Model 2 (elasticity)** does not rely on reform timing at all:

- It uses within-terminal variation in K/L over time, net of common shocks, to estimate the elasticity of LP w.r.t. K/L.
- This is interpreted as a production-function parameter, cross-checked against capital-income shares.

6.2 What we do *not* claim

- We **do not claim** that K/L is exogenous or that engineering projects are valid instruments.
- We **do not implement IV-based causal mediation** in the Dippel–Ferrara sense.
- Instead, we present the mediation step as **mechanism accounting**: given an empirically grounded elasticity, how much of the reform-induced LP change can reasonably be ascribed to observed capital deepening?

6.3 Planned robustness checks

To address advisor feedback and remaining concerns:

1. Alternative labor measures

2. Re-estimate all models with different L series (hours, headcount, wage-based) and corresponding LP and K/L definitions.

3. Alternative K series

4. Use several Path B variants (different depreciation schedules, different treatment of revaluations or disputed projects).
5. Possibly include a "no-big-project" variant as a lower bound.

6. Stacked event-studies

7. Implement stacked NYT-style event-studies when pooling treatments or dealing with multiple reforms in close succession.

8. Shock controls

9. Include dummies for major exogenous events (wars, COVID) in all specifications as additional robustness.

7. Implications for Rewriting the Thesis

7.1 Abstract and contribution

The abstract should be rewritten to highlight:

- Causal estimates of the effect of reforms on terminal-level LP (Model 1A).
- Parallel estimates of the effect on K/L (Model 1B), based on newly constructed monthly capital series.

- A mechanism analysis that uses a theory-consistent elasticity of LP w.r.t. K/L (Model 2) to gauge the contribution of capital deepening to observed productivity gains.
- Emphasis on **measured capital dynamics and service-quality outcomes**, not IV-based mediation.

7.2 Empirical design section

The design section should be reorganized into four subsections:

1. **4.1 NYT event-study for LP (Model 1A).**
2. **4.2 NYT event-study for K/L (Model 1B).**
3. **4.3 Elasticity of LP w.r.t. K/L (Model 2).**
4. **4.4 Mediation-style decomposition and robustness.**

Language about IV mediation should be removed or relegated to a short subsection explaining why IV was considered but ultimately not pursued.

7.3 Data and measurement section

The data section should:

- Expand on the **K construction** (Path B, big projects, deflator), making it a core contribution.
- Clarify the construction of LP and L, including the transition from proxy-based LP to actual hours.
- Explain how LP, K, and L are combined to form K/L and mapped onto terminals.

7.4 Results section structure

The main results chapter can be organized as:

1. **Figure 1 / Table 1:** Descriptive trends in throughput, LP, and K/L.
2. **Figure 2 / Table 2:** Event-study plots for LP (Model 1A) by reform.
3. **Figure 3 / Table 3:** Event-study plots for K/L (Model 1B) by reform.
4. **Table 4:** Elasticity estimates $\hat{\lambda}$ under different K/L and L specifications (Model 2).
5. **Table 5:** Mediation summary: for each reform and horizon, total LP effect, K/L effect, implied indirect component, residual component.
6. **Appendix:** Alternative specifications (stacked event-studies, shock controls, alternative K/L definitions).

8. Practical Roadmap

To implement this redesigned thesis, the remaining work can be grouped into three streams:

8.1 Data work

- Finalize terminal-level LP panel using true monthly labor hours when available, plus proxy versions.
- Complete the Path B capital series and its variants for Haifa legacy (and to the extent possible, for other terminals).

- Build the merged terminal×month dataset with LP, K/L, reform clocks, and shocks.

8.2 Estimation work

- Estimate Model 1A and 1B for all reforms, including stacked versions.
- Estimate Model 2 (elasticity) and compute $\hat{\lambda}$ and its robustness variants.
- Carry out the mediation-style decomposition for key horizons and report results.

8.3 Writing work

- Rewrite the proposal text into thesis chapters, reflecting the new econometric design and capital measurement strategy.
- Integrate the new models, tables, and figures into the methods and results chapters.
- Ensure conceptual consistency: K/L as a measured outcome and mediator, not as an instrumented regressor.

This redesigned plan preserves the original spirit of the project—evaluating the effects of Israel’s port reforms on productivity and capital deepening—while bringing the econometric design in line with the data, the feasibility of identification, and the substantial progress you have made on measuring capital at the port-terminal level.